SYMATICS

Gen1 POC & FPGA Proof-of-Concept Plan

From architecture validation to multi-node FPGA proof-of-concept — one page, one timeline.

Working draft · 14 October 2026 · supersedes the December-first-results framing

1. Milestones

When	Milestone	Detail / exit criterion
Now → 19 Oct	SIF-Lite fabric drop (Wolley)	SIF-Lite sled firmware & flit path delivered for bench bring-up
20 Oct	MPU-Lite compute cards (ePIC)	2 × V80-class MPU-Lite nodes received; VPK120 SIF-Lite nodes on hand
26–30 Oct	Integration week	Two-node path live on the bench: host image load, one layer executes, state crosses V80 → SIF-Lite → fiber → SIF-Lite → V80
30 Oct	**Go / no-go — architecture simulation & validation POC complete (end of October)**	Identical cycle count every run; spool swap 10/50/100/500 m shifts latency by the predicted amount; no jitter, no retry
November	**Multi-node FPGA proof-of-concept build — complete end of November**	Hardened two-node platform; measurement harness; reproducible runs with logged configuration
December	First results published as measured	Fixed-latency state transfer between nodes; latency-determinism and state-residency measurements, each with its configuration
2027	Scale-out	64-chip domain equivalent; broader workload characterization; workload-level performance & economics

2. What the POC must demonstrate

Topology and timing — the two claims the architecture rests on: (a) state moves between nodes on a compile-time schedule with no runtime routing; (b) latency is pipe depth, so it changes only by the amount a physical change (fiber length) predicts. Everything else — throughput at workload level, power, economics — is quantified later and published with its configuration.

3. Exhibit gates (Singapore, 5–7 Nov)

Each level ships independently; the exhibit is presentable at every gate.

Gate	What shows	Status
L0+	Lit static exhibit — LED pulse traces MPU → SIF → fiber → MPU; spool switch lengthens the segment; fault button shows the red poison pulse	Guaranteed floor — built from 3 Oct
L1	Live screen fed by the bench counter	Needs only network to the bench
L2	On-site live — two FPGA nodes joined by real fiber over the booth link	Target; follows the 30 Oct go/no-go

4. Ownership

Workstream	Owner
Rack & exhibit build (24U rack, sleds, LED controller, fiber looms, screen, freight)	Symatics Taiwan — PM + part-timer
SIF-Lite fabric & integration support	Wolley
MPU-Lite compute cards	ePIC
POC compiler, state-path scheduling, measurement harness	Symatics engineering (Wayne Wu, CTO)

5. Out of scope — for this phase and the booth

Optically switched links (OCS) · multi-lane · FEC · epoch sync across nodes · any tokens/sec or TCO figure on placards or in press. Say: FPGA-based validation platform; deterministic state movement; optical link; evidence-building programme. Never say: finished AI chip, commercial silicon performance, guaranteed power or TCO savings, “world’s first”.

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